MILITARY SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, HEAVY DUTY, 2 AMPERE-HOURS, SEALED CELLS, 26 VOLTS WITH 115V 400 HZ INTEGRAL CHARGER

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the storage battery described herein shall consist of this document and the latest issue of Specification MIL-B-8565.

REQUIREMENTS:

1. Dimensions and configuration. The dimensions of the battery shall be as shown on Figure 1. All dimensions are in millimeters. All recurrent dimensions are typical.

2. Military part number. The military part number of the battery shall be D8565/1-1.

3. Electric connections. Each battery shall include as its electric connections an MS3112E14-5P receptacle which shall be installed on the battery in the location and orientation shown on Figure 1, and shall be connected as follows:

   (A) Pin C shall be battery negative, and pin A shall be battery positive.

   (B) Input for trickle charging shall be on pins D and C. Pin C shall be input ground.

4. Trickle charging circuit. The trickle charging circuit shall keep the battery fully charged for 30 days without damage or degradation of performance when fully charged initially and also shall be capable of fully charging the battery from any state of discharge in a maximum of 12 hours under the following conditions:

FSC 6140

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Temperature Range	-30° to 50°C
Input Voltage	108-118 VAC
Input Frequency	380-420 Hz
Input Current	1 ampere maximum

5. Nominal potential. The nominal potential of the battery shall be 26 volts.

6. Nominal capacity. The nominal capacity of the battery shall be 2 ampere-hours. (2.0AH/1-HR/26.7°C/21 Volts)

7. Marking. If the identification marking and instructions are on a removable cover, the manufacturer's name and part number shall also be marked in the location and orientation shown on Figure 1.

8. Mass. The mass of the battery shall not exceed 3.9 kilograms in any state of charge.

9. Specification. The batteries shall conform to requirements of MIL-B-8565 for no-maintenance, heavy duty, low discharge rate batteries except as modified by the following:

GENERAL:

A. Replace "procurement standard", wherever it occurs, with "specification sheet".

B. Replace "design standard", wherever it occurs, with "specification sheet".

C. Requirements for receptacle-connected batteries are not applicable to this battery.

4.6.8 through 4.6.8.2 Charges. Replace with:

4.6.8 Charges. After each discharge, except those of the Life Test, the battery shall be allowed to remain on open circuit in an ambient of 21.1° ±5°C for 1.0 ±0.1 hour, and shall then be charged for a period not to exceed 12 hours in an ambient of 21.1° ±5°C as specified below.

4.6.8.1 For the life test. After each discharge of the Life Test, the battery or cell shall be given a charge during which the current shall be maintained at one value for the first 2.0 ±0.5 hours of the charge, and at the same or another value for the rest of the charge.

The government testing agency will decide what currents to use. The current during each part of the charge shall be within ±25 percent of the

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nominal value at all times. The charge put into the battery shall be within ±10 percent of the sum of the products of the nominal currents and the respective nominal times.

4.6.8.2 For all other tests. After each discharge other than those of the Life Test, the battery shall be left idle at 21.1° ±5°C for 1 ±0.1 hour, and shall then be charged using the integrated charger for a period not to exceed 12 hours or shall be constant current charged at the c/10 rate for a period of 12 ±0.1 hours.

4.7.2 Dielectric strength. Delete.

4.7.13 Life. The number of life cycles shall be 25.

4.7.15 3-Minute discharge. Delete.

4.7.25.1 Special tests: Add the following:

4.7.25.1 Step discharge. The battery shall be subjected to a step discharge at -23.3° ±2°C. The current shall be maintained at the value listed below for each period listed below during the discharge. The potential during each period of the discharge shall not be less than the value listed below.

Period of time	First 1 minute	Next 5 seconds	Next 30 minutes
Current, Amperes	1.8	15.0	1.8
Minimum Potential, Volts	21	18	20

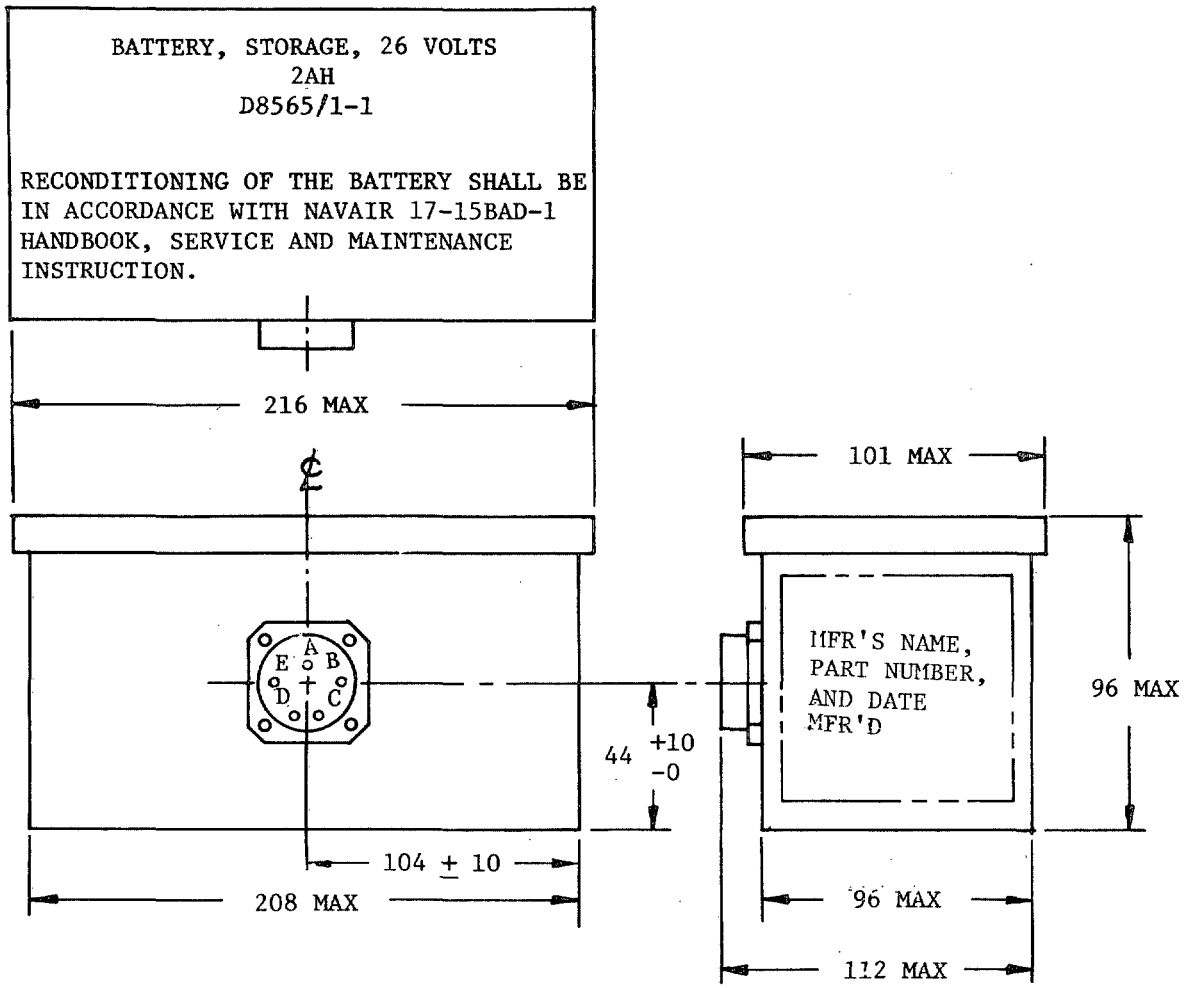
4.7.25.2 Trickle charge. This test shall be conducted at 21.1° ±5°C. A completely discharged battery shall be trickle charged for 12 hours. The battery shall then be subjected to a 2-ampere capacity discharge test to 21 volts. The battery shall be charged in accordance with 4.6.8 and then stored for 30 ±1 days during which time it shall be subjected to a trickle charge for 1 ±.05 hour/day. At the end of the 30-day period, the battery shall be subjected to a 2-ampere capacity discharge test to 21 volts. The final capacity shall not be less than 95 percent of the previous capacity discharge or 2 ah, whichever is greater.

4.7.28 Fungus. Delete

4.7.29 Humidity. Add: After humidity test, charge the battery for 12 hours in an ambient of 21.1° ±5°C using the integral charger. The battery shall then be subjected to a 1-hour discharge.

4.7.30 Salt fog. Replace the text with: With the cover secure on the battery, it shall be subjected to the salt fog test of Procedure I, Method 509.1 of MIL-STD-810C.

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PART NO.
D8565/1-1

FIGURE 1. Battery configuration.

Preparing activity:
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(Project No. 6140-N433)

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OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE DOD-B-8565/1(AS) BATTERY, STORAGE, AIRCRAFT, HEAVY DUTY, 2 AMPERE-HOURS, SEALED CELLS, 26 VOLTS WITH 115V 400 HZ INTEGRAL CHARGER

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

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DATE

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